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# Fast displacement and structural transition of cyclohexanethiol self-assembled monolayers by octanethiols on Au (111)

Seungwook Kwon, Youngsik Choi, Jungseok Choi, Hungu Kang, Hoeil Chung, Jaegeun Noh *

*Department of Chemistry, Hanyang University, 17 Haengdang-dong, Seongdong-gu, Seoul 133-791, Republic of Korea*

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ABSTRACT

Displacement of cyclohexanethiols (CHTs) self-assembled monolayers (SAMs) by octanethiols on Au (111) (OTs) was examined by scanning tunneling microscopy (STM) and contact angle measurements. We revealed that the fast displacement of CHT by OT takes place within a few minutes, and then displacement proceeds slowly to form the closely packed OT SAMs. The main driving force for the unusual fast displacement is due to the large increase of chemical interactions between the sulfur and gold atoms, and the van der Waals interactions between alkyl chains after displacement of CHT by OT. STM imaging clearly demonstrated the structural transitions from the $(5\times2\sqrt{10})R48^\circ$ structure of CHT SAMs to the $(\sqrt{3}\times\sqrt{3})R30^\circ$ or $c(4\times2)$ structures of OT SAMs via an intermediate phase that were often observed during alkanethiol SAM growth on gold. In this study, we found that CHT SAMs can be used as a new transient layer for the fabrication of nanostructures on a surface.

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## 1. Introduction

Molecular self-assembly has been used as a principle technique in nanopatterning and in the fabrication of nanoscale molecular assemblies on solid surfaces for many technical applications [1–4]. It has generally been believed that self-assembled monolayers (SAMs) are excellent molecular systems for tuning the physical and chemical properties of a surface using functionalized organosulfur molecules [1–8]. The formation of well-ordered and stable SAMs on gold is driven mainly by the chemical interactions between the sulfur and gold atoms, and the van der Waals interactions between molecular backbones.

It has been reported that SAMs with a low molecular density and weak lateral interaction can be used as temporary protective layers for tailoring nanoscale surface properties and in surface patterning, because they are removed easily by dipping them in solutions containing target compounds [9–11]. SAMs on gold can be fabricated by using thiols with the spherical hydrocarbon cage tail groups [9–12]. Scanning tunneling microscopy (STM) imaging revealed that 1-adamantanethiols with a large spherical cage on Au(111) form well-ordered SAMs with hexagonal packing arrangements in the unit cell, which are described as the $(7\times7)$ and $(7\times7)R21.8^\circ$ structures corresponding to less dense phases than that of the $(\sqrt{3}\times\sqrt{3})R30^\circ$ structure for conventional alkanethiols [11]. The nanoscale binary SAMs and patterned surfaces were successfully prepared using the SAM systems and the molecular exchange method [9,10].

A number of molecular exchange experiments have been performed using SAMs to understand exchange kinetics and whether mixed or phase-separated domains are formed after exchange [13–22]. Short-chain alkanethiols chemisorbed on gold can be replaced by long-chain alkanethiols in solution driven by the energy gained resulting from an increase of van der Waals interactions between longer alkyl chains [14]. It was reported that the replacement of di(phenylethynyl)benzenethiols by decanethiols on Au(111) proceeds very slowly for a few weeks. On the other hand, the replacement of propanethiol by *p*-benzenedimethanethiol on Pt(111) is nearly accomplished after 42 h, resulting in the formation of ordered SAMs with an upward orientation [21]. A recent kinetic study revealed that the replacement of adamantanethiols by dodecanethiols on Au(111) occurred unusually fast and finished after 2 h, and was driven by the enthalpic contributions resulting from an increase in the chemical interactions between the sulfur and gold atoms, and of the van der Waals interactions between alkyl chains [22]. It was also observed that the molecular replacement initiates quickly from domain boundaries and vacancy islands (VIs), and subsequently a much slower replacement occurred in the ordered domains of SAMs [17–20].

Recently, our STM study elucidated that the adsorption of cyclohexanethiols containing a cyclic ring on Au(111) leads to the formation of two-dimensional (2D) ordered phases having molecular density lower than that of alkanethiols [23,24]. It was also expected that the van der Waals interactions between aliphatic cyclic rings are very weak, due to a flexible ring

* Corresponding author. Tel.: +82 2 2220 0938; fax: +82 2 2299 0762.
*E-mail address:* jgnoh@hanyang.ac.kr (J. Noh).

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conversion in CHT SAMs. From these structural and interaction viewpoints of CHT SAMs, we expect that CHT SAMs on gold can be used as a new transient layer in the fabrication of nanostructures and control of the 2D structure of target molecules on the surfaces using the displacement method. The main purpose of this paper is to elucidate the displacement processes and kinetics of CHT by alkanethiols on Au(111) using STM and contact angle measurements. Here we report the first molecular-scale features concerning the displacement processes of CHT molecules by octanethiols on Au(111) and their unusually fast molecular exchange in solution.

## 2. Experimental

Cyclohexanethiol ($C_6H_{11}SH$, CHT) and octanethiol ($C_8H_{17}SH$, OT) were purchased from Tokyo Chemical Industry (Tokyo, Japan) and used without further purification. The Au(111) substrates were prepared by thermal evaporation of gold onto freshly cleaved mica sheets under a pressure of $10^{-7}$–$10^{-8}$ Torr, as described in a previous paper [6].

Pre-covered CHT SAMs were formed by immersing the Au(111) substrates in a 1 mM ethanolic solution of CHT for one day. Displaced OT SAMs were prepared by immersing the pre-covered CHT samples in a 1 mM ethanolic solution of OT for the specified time period. All SAM samples were rinsed in pure ethanol and blown dry with nitrogen prior to surface analysis.

STM measurements were carried out under ambient conditions using Nanoscope E (Veeco, Santa Barbara, CA) with a commercially available Pt/Ir tip (80:20). All STM images were obtained using the constant current mode. A bias voltage ranging from 400 to 600 mV and a tunneling current ranging from 0.3 to 0.5 nm were applied between the tip and sample. Contact angle measurements were performed by using an automatic contact angle measuring system (model: Easydrop DSA20S, KRÜSS, Germany). Water contact angles on the SAM samples were measured at room temperature using the sessile drop method and obtained by averaging the results of five droplets.

## 3. Results and discussion

STM images in Fig. 1 clearly show the formation of well-ordered CHT SAMs on Au(111) after 1-day deposition in a 1 mM ethanol solution at room temperature. The STM image in Fig. 1a depicts the typical molecular features of CHT SAMs, such as the large ordered domains with three domain orientations, domain boundaries (dark arrows), vacancy islands (VIs), and the domain dislocations (white arrows). Compared to alkanethiols SAMs on Au(111), CHT SAMs have wider domain boundaries consisting of disordered phases. This difference can be attributed to the conformational dynamics of cyclic rings, because CHT molecules at the domain boundary regions have larger intermolecular spaces for the conformational change of cyclic rings. The molecularly resolved STM image in Fig. 1b shows an oblique unit cell containing two different molecular spots and assigned a $(5 \times 2\sqrt{10})R48^\circ$ structure. It is expected that this unique structure for CHT SAMs is due to the existence of two conformers: the axial and equatorial conformers. The average areal density of CHT adsorbed on gold is calculated to be 87.8 $Å^2$/molecule, whereas that of alkanethiol for the $(\sqrt{3} \times \sqrt{3})\ R30^\circ$ structure is calculated to be 21.6 $Å^2$/molecule. It is therefore clear that the molecular density of CHT SAMs on Au(111) is approximately one-fourth of alkanethiols SAMs. We assumed that such a lower molecular density results from the larger molecular volume of CHT molecules due to their ring structures and flexibility. Structural details of CHT SAMs on Au(111) were described in our previous communication paper [23]. Based on these structural features, we expect that CHT molecules on gold can be easily replaced by other target molecules, and CHT SAMs can be used as a new transient layer in the fabrication of nanostructures on a surface, as in the case of adamantanethiol SAMs.

Fig. 2 shows the variation of the water contact angle of SAMs on Au(111) after displacement of CHT molecules by OT molecules on Au(111) with displacement time: (a) as-deposited CHT (single component), (b) 3 s, (c) 2 min, (d) 5 min, and (e) 20 min. Water contact angle increases with displacement time, implying that the formation of SAMs with the higher hydrophobicity results from the displacement of CHT by OT. It has been reported that alkanethiol SAMs with methyl-terminal groups on gold have a large contact angle, ~105°, which strongly implies the formation of well-ordered and highly hydrophobic SAMs [16]. We found that the fast displacement of CHT by OT takes place within several minutes, and then the displacement proceeds slowly to form closely packed OT SAMs. A contact angle of 101° was observed after displacement for 20 min, suggesting the formation of OT SAMs as a result of displacement of CHT by OT on Au(111). This unusual fast displacement can be attributed to the low molecular packing density of CHT SAMs, as revealed by our STM result. Therefore, CHT molecules on Au(111) can be easily displaced by

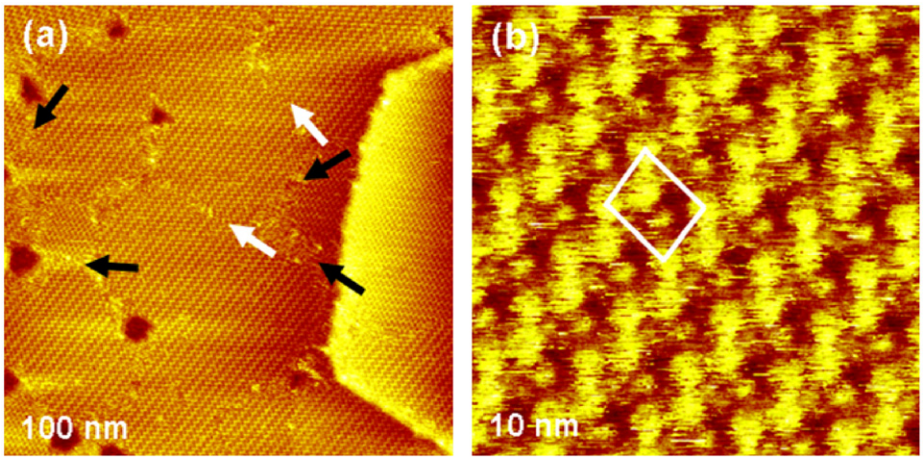


**Fig. 1.** STM images of CHT SAMs on Au(111) prepared after the immersion of Au(111) substrates in a 1 mM ethanol solution of CHT for one day. (a) STM image of CHT SAMs showing long-range 2D ordered domains consisting of domain boundaries and VIs. (b) High-resolution STM image showing the 2D molecular arrangements of CHT SAMs.

OT molecules to form the $(\sqrt{3}\times\sqrt{3})R30^\circ$ or $c(4\times2)$ structures with a higher molecular density. As a result, the interactions between the sulfur and gold atoms can be greatly increased, as can the van der Waals interactions between octyl chains. This should be a key driving force for the fast displacement. We found that the displacement kinetics of CHT SAMs (87.8 Å$^2$/molecule) by alkanethiols is much faster than those of adamantanethiol SAMs (38.9 Å$^2$/molecule) [22]. The large difference in kinetics can be ascribed to a difference in the molecular packing density. The energy gain for CHT SAMs is thus much larger than that for adamantanethiol SAMs, because CHT SAMs have more empty space on the surface to form gold–sulfur chemical bonds via displacement.

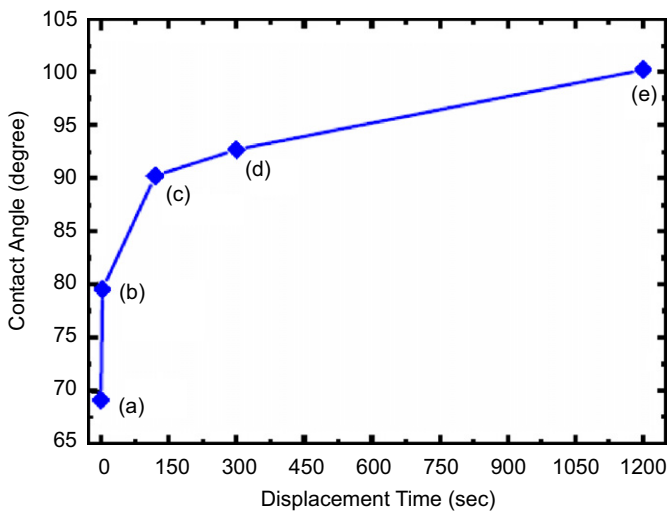


**Fig. 2.** Water static contact angles of SAMs on Au(111) after displacement of CHT SAMs by OT as a function of time. The displacement time was (a) as-deposited CHT, (b) 3 s, (c) 2 min, (d) 5 min, and (e) 20 min.

To reveal the molecular-scale features with displacement time, we examined the structures of SAMs on Au(111) by STM as a function of displacement time. Images in Fig. 3a and b clearly show significant structural changes of CHT SAMs on Au(111), even after displacement for 3 s. The structural characteristics of CHT SAMs as shown in Fig. 1 were totally changed. New structural features, such as striped phases with low surface coverage (region A), disordered phases (region B), and well-ordered phases of type $(\sqrt{3}\times\sqrt{3})R30^\circ$ (regions C) appeared. The striped and disordered phases were usually observed as intermediate phases during alkanethiol SAM growth prior to the formation of densely packed SAMs having the $(\sqrt{3}\times\sqrt{3})R30^\circ$ or $c(4\times2)$ structures [25,26]. This STM result strongly suggests that most CHT molecules are already replaced by OT molecules after even a very short displacement time of only 3 s. As the displacement time increases to 2 min, the area of the striped phases decreases, and the area of the well-ordered phase with $c(4\times2)$ structures increases (Fig. 3c and d). We note that the transitions from the $(\sqrt{3}\times\sqrt{3})R30^\circ$ to the $c(4\times2)$ structures occurs as a result of the structural rearrangement of densely packed SAMs with time evolution to form a more stable packing structure [27]. We observed the same structural transitions occurred in the displaced SAM samples (see Fig. 3b and d). After displacement for 5 min, the striped phases completely disappeared and the well-ordered domains (with uniform brightness) were formed due to structural rearrangements to maximize the van der Waals interactions between alkyl chains (Fig. 3e and f).

On the other hand, the surface structures of SAMs on Au(111) formed after displacement for 20 min (Fig. 4) are nearly the same as those formed after displacement for 5 min. The area of disordered phases, however, decreased after longer displacement, as shown in Fig. 4b. Here we note that the structures observed in Fig. 4 are similar to those observed from fully covered OT SAMs

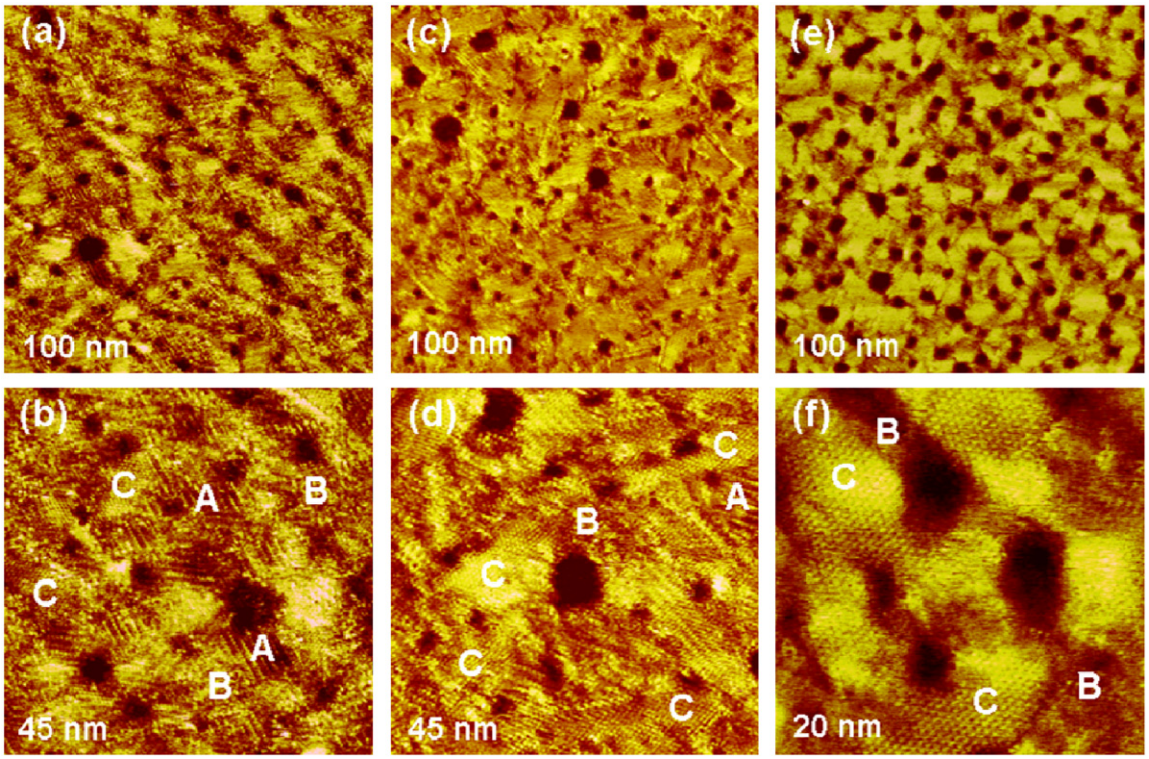


**Fig. 3.** STM images of SAMs on Au(111) formed after displacement of CHT by OT with displacement time: (a),(b) 3 s, (c),(d) 2 min, and (e),(f) 5 min.

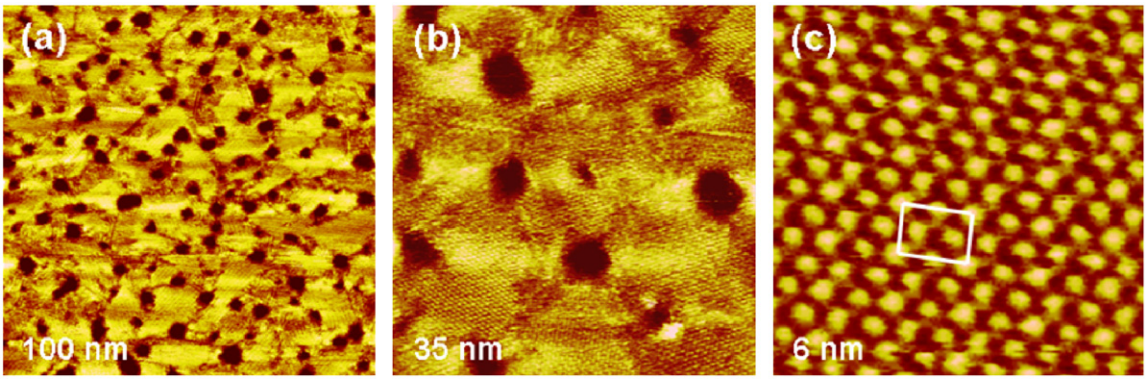


**Fig. 4.** STM images of SAMs on Au(111) formed after displacement of CHT by OT for 20 min.

formed after 14 h immersion in a 2 mM solution of OT [28]. This result means that CHT molecules can be totally replaced by OT within 20 min. The high-resolution STM image in Fig. 4c obviously shows the $c(4\times2)$ structure, which can be observed from a fully covered OT SAMs on Au(111). This is a strong evidence for the displacement of CHT by OT molecule. From our STM study, we clearly demonstrated the structural transition from the $(5\times2\sqrt{10})R48^\circ$ structure of CHT SAMs to the $(\sqrt{3}\times\sqrt{3})R30^\circ$ or $c(4\times2)$ structures of OT SAMs as a function of displacement time, which is consistent with the results obtained from the contact angle measurements that show the increase of contact value with displacement time.

## 4. Conclusions

From STM and contact angle measurements, we demonstrated that CHT molecules on Au(111) can be easily exchanged by OT molecules within a few minutes. As a result, we believe that CHT SAMs can be used as a new transient layer in the fabrication of nanostructures on a surface. The driving force for the unusually fast displacement is due to a large increase of interactions between the sulfur and gold atoms, and of the van der Waals interactions between octyl chains after displacement of CHT by OT. In addition, STM imaging clearly revealed the structural transition from the $(5\times2\sqrt{10})R48^\circ$ structure of CHT SAMs to the $(\sqrt{3}\times\sqrt{3})R30^\circ$ or $c(4\times2)$ structures of OT SAMs via an intermediate phase as the result of molecular exchange.

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